Project Proposal

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The theme I have chosen to create a project for is Mobility Innovation. My application will help people locate a city bike and show whether it is available to use. The bike options will be a normal bicycle, or an electric bicycle based on user input. The app will allow for adding a bike station, the number of bikes available per station and whether it is electric or not. Prices will differ; a normal bike will be 1 euro per KM, and an electric will be 1.50 per KM. The user will be asked how far they wish to travel and must insert a city bike station close to their destination.

The data structures I plan to use are Queue, Stack and Single Linked list. I will implement a queue to keep track of the users waiting to use a bike at a station. Stack will be used to monitor bikes which have been returned recently. Single-linked list will be implemented to show stations in the area. Given that a queue is a First-In-First-Out abstract data type, its functions will suit my queue of users waiting to rent a bike. Stack is a Last-In-First-Out ADT, which can work well to monitor the return of bikes to a station. Single linked list will be useful to show the chain of bike stations in the area.

The theme of this project is to improve mobility in the area, and I believe an app like this will be beneficial to people who are travelling to and from work, college and school. This would also have a positive impact on the economy and traffic in the city as fewer people are driving or taking buses and trains, therefore lowering congestion and emissions.

I will develop a user-friendly graphical interface that is easy to read and use on the go. When developed, this could be used on mobile devices to assist people in their everyday journey. A GUI that is easy to read and intuitive is key in this situation, as people do not have time to try to navigate menus or scroll through pages to find the correct bike/station for them. Efficiency and usability are at the forefront of this application.

The app will allow for user input, taking in object data and storing it in each ADT. The user will input the station required, the bike required, the journey length and whether the bike is returned. These inputs will all be stored using a queue, a stack, and a Single-linked list. Two CRUD operations will also be integrated into this app to allow the user to read the stations. They will traverse the linked list to find the correct station for them and select it.

To summarise this application, the user will be able to find bikes in the area based on bike stations, pick a bike, start a journey and end it at a station. It will show available bikes and stations nearby. Finally, it displays a message with the start, end, cost and bike type. I will attach my class diagram to showcase the application's overall layout further.

Link to GitHub Repository: https://github.com/RW0990/DataStructuresCA1RyanWhite.git

Class Diagram